Research activity

A limit is not a last term — 30 min.

Consider the sequence (u_n) defined on $\mathbb{N}$ by $u_n = 2 + \frac{3}{n+1}$.

1. Calculate u_0, u_9 and u_{99}. Is there an integer n such that $u_n = 2$? Justify.
2. Determine the smallest integer N such that, for every integer $n \geq N$, $|u_n - 2| < 0.01$.
3. Let $\varepsilon > 0$. Give, in terms of ε, an integer N such that $|u_n - 2| < \varepsilon$ for every $n \geq N$.
4. Define on $\mathbb{N}$ the sequence (v_n) by $v_n = 2 + \frac{(-1)^n}{n+1}$. Prove that it converges to 2 and that it is not monotonic. What does this example show?